

Supply Chain & Inventory Performance Analytics

This report condenses a data analytics project that transforms operational transaction data into actionable inventory and supply chain insights. The analysis covers demand patterns, inventory levels, warehouse capacity, supplier performance, transportation cost, order accuracy and backorder activity. The final deliverable is an interactive Power BI dashboard presenting KPIs and trends for inventory planning and operational monitoring.





Project Overview & Business Objectives

Measure Core Metrics

Track total sales volume, inventory levels and transportation expenditure to inform strategic decisions.

Compare Demand

Analyze demand by region and product category to prioritize replenishment and allocation.

Inventory Distribution

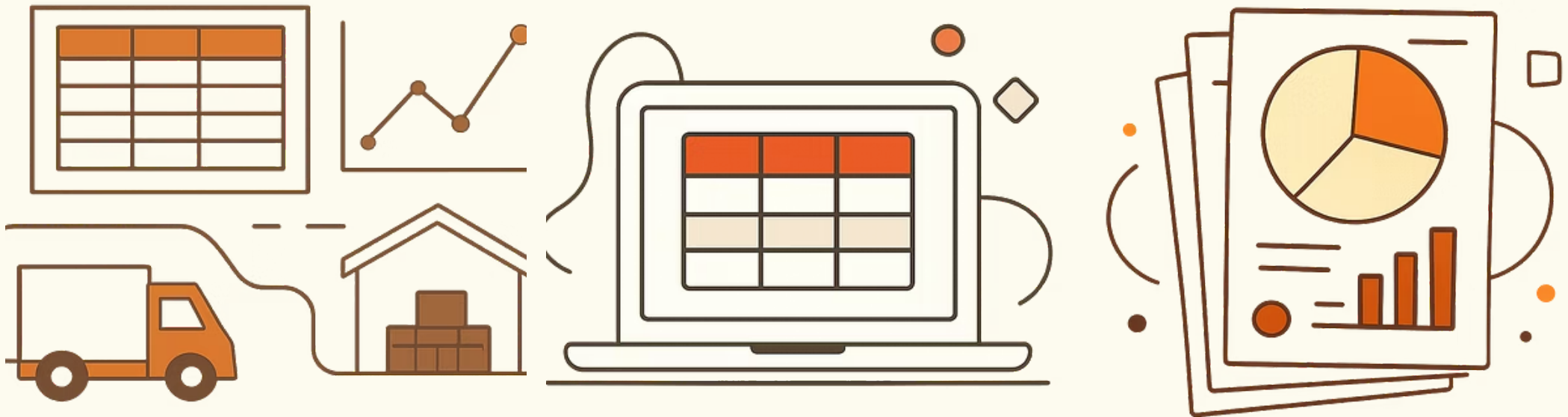
Understand inventory across warehouses to reveal concentration and capacity issues.

Interactive Dashboard

Deliver a Power BI dashboard for KPI tracking and operational monitoring.

Dataset Summary

The dataset contains 1,200 rows and 15 columns and includes date, region, category, supplier, warehouse, order status and operational measures. This structured transactional table forms the foundation for exploratory analysis, SQL aggregation and the Power BI visuals.



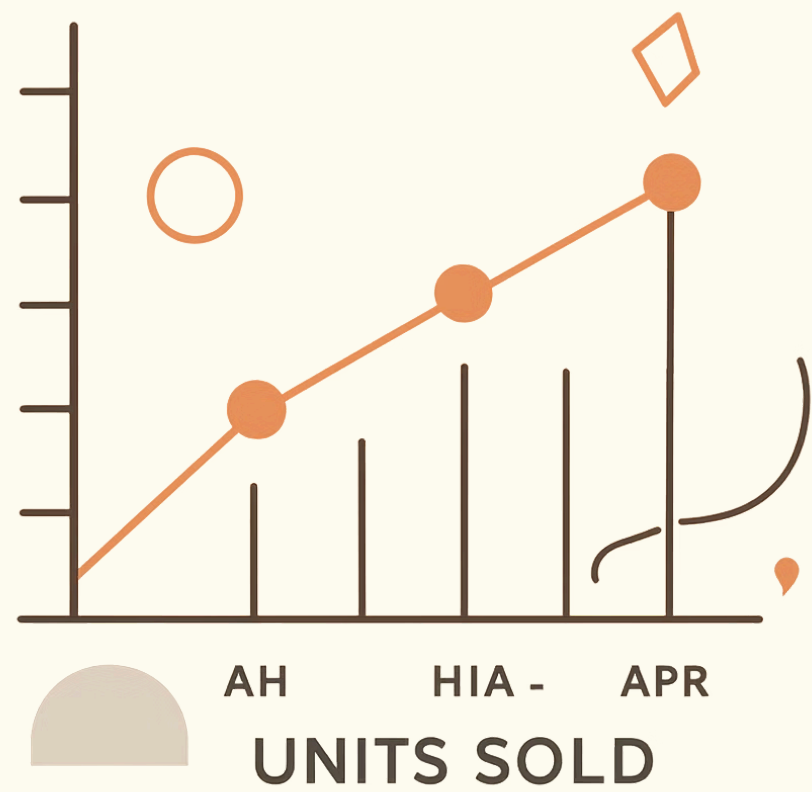
Exploratory Data Analysis (Python)

Python (pandas) was used to load, inspect and explore the dataset prior to SQL and dashboard work. EDA steps included descriptive statistics, validation of boolean flags (Order Accuracy, Backorder), and investigation of relationships among demand, inventory, suppliers and operational measures. Outputs fed the SQL layer and Power BI aggregates.



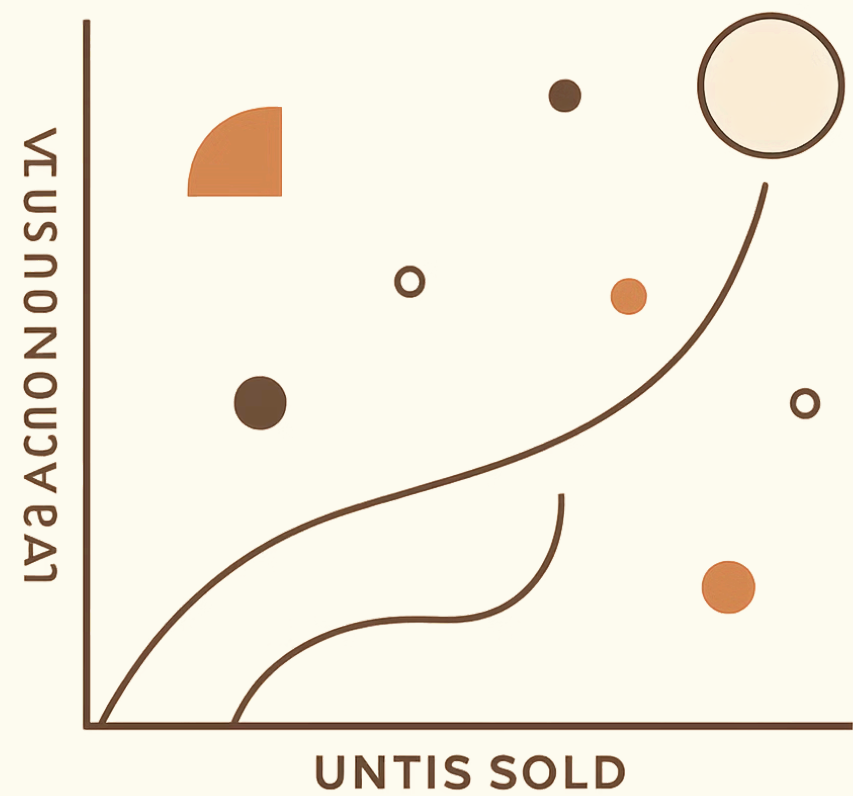
Key EDA Visuals

Monthly Units Sold



Trend analysis highlights seasonality and peak months for demand.

Inventory vs Units Sold



Scatter shows the relationship between stock on hand and realized sales, informing replenishment thresholds.

SQL Analysis & Techniques

PostgreSQL hosted the clean supply_chain table and performed KPI calculations and grouped analysis. Typical SQL constructs used: SUM, AVG, COUNT, CASE, GROUP BY and ORDER BY to convert transactional rows into business-level metrics such as units sold by region, inventory by warehouse, average lead time, backorders and order accuracy by supplier.



Table Creation & Validation

Created and validated the supply_chain table, ensuring data integrity for downstream analysis.



KPI Calculations

Computed aggregates for units, inventory, lead times, backorders and transportation cost.



Power BI Dashboard — Structure

The Power BI report is organized into three pages: Executive Overview (performance KPIs and trends), Inventory & Warehouse (capacity, inventory distribution, backorders) and Supplier Performance (lead time, accuracy, backorders, transportation cost). Slicers enable filtering by region, category, supplier, warehouse and order status.



Power BI Pages — Visual Highlights



Page 1 — Executive Overview

KPI cards: Total Units Sold, Total Inventory, Total Transportation Cost, Average Lead Time, Backorder Rate. Includes monthly trend and breakdowns by region and category.



Page 2 — Inventory & Warehouse

Capacity utilization, inventory by warehouse and category, inventory vs units sold scatter, and backorders by warehouse for operational prioritization.



Page 3 — Supplier Performance

KPIs include average lead time, order accuracy, backorders and transportation cost; visuals show per-supplier lead time, accuracy and cost comparisons.

Business Insights — Measured Outcomes

Demand & Category: Total demand is 640,671 units. North leads with 170,175 units; South has 148,997. Clothing and Electronics record 168,642 and 168,532 units respectively.

Inventory & Warehouse: Total inventory is 3,036,568 units. Warehouse 2 holds 1,030,158 units. Dashboard pairs inventory with demand to contextualize stock levels.



Supplier Metrics, Risk & Recommendations

Supplier Performance & Risk: Overall average lead time is 15.74 days; Supplier B averages 16.17 days. Supplier D has the most backorder records (37). Overall order accuracy is 91.33% (supplier range 89.45%–92.31%). Total transportation cost is 9,268,222.19 and overall backorder rate is 9.67%.

1. Strengthen Inventory Planning

Use regional/category demand with warehouse levels to optimize replenishment and reduce excess or shortage risk.

2. Monitor Lead Time

Track supplier lead times and investigate consistently high values, especially where they coincide with backorders.

3. Reduce Backorder Risk

Use backorders by supplier and warehouse views to identify recurring issues and prioritize operational reviews.

4. Improve Order Accuracy

Compare supplier accuracy periodically; review lower performers for process or fulfillment remediation.

5. Control Transportation Cost

Monitor transportation by supplier and region and combine cost with demand/service metrics to uncover efficiency opportunities.